



Adversarial Motion Planning

ENAE644 Final Project

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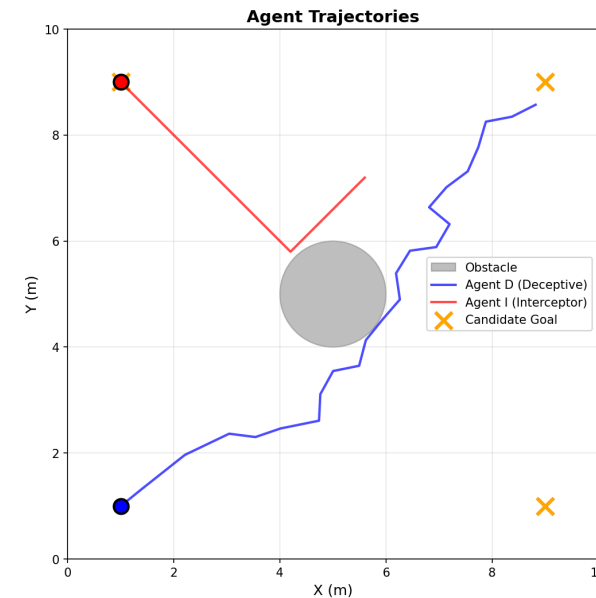
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1. Problem Statement

1.1 A Two-Agent Adversarial Scenario

- Two competing planners share a continuous 2D workspace.
- **Agent D** (deceptive) — must reach a **hidden** goal while concealing its intent from any observer.
- **Agent I** (interceptor) — must **infer** Agent D's goal and **intercept** before D arrives.
- Both planners go beyond plain RRT*: they layer learning- and game-theoretic components on top of sampling-based search.
- Goal: a **direct adversarial pitting** — one deceiver vs. one learning interceptor in the same sandbox.



2. Motivation & Related Work

2.1 Why Adversarial Motion Planning?

Applications. Game AI, privacy-preserving navigation, autonomous-vehicle interaction, defense and security.

Deceptive planning

- Nichols et al., 2022 — **Adversarial RRT*** with a learned observer.
- Xu et al., 2020 — MILP for goal-ambiguity maximization.

Game-theoretic / interactive planning

- LucidGames (Le Cleac'h et al., 2021).
- Espinoza et al., 2022 — game-theoretic MPC + neural prediction.

Gap addressed. Direct head-to-head evaluation of a **deceptive** planner against a **learning** interceptor in a strategy-game-style setting.

Goal recognition + interception

- Tastan et al., 2012 — IRL + particle filter (FPS games).
- Zeng & Xu, 2023 — IRL + game-theoretic counter-deception.

Adversary-aware

- Netter & Vamvoudakis, 2024 — GP-classifier-based adversary avoidance.

3. Technical Formulation

3.1 Workspace and Agents

Workspace. Bounded 2D region $\mathcal{W} \subset \mathbb{R}^2$, static obstacles $\mathcal{O} \subset \mathcal{W}$, free space $\mathcal{W}_{\text{free}} = \mathcal{W} \setminus \mathcal{O}$.

Agents. Positions $x_D(t), x_I(t) \in \mathcal{W}_{\text{free}}$; trajectories $\xi_D, \xi_I : [0, T] \rightarrow \mathcal{W}_{\text{free}}$ subject to kinodynamic constraints $\xi \in \Xi$.

Goals.

- $\mathcal{G}_D = \{g_1, \dots, g_M\}$ – **candidate** goals, known to both agents.
- $g^* \in \mathcal{G}_D$ – the **true** goal, known **only** to Agent D.

Belief. (Agent I's distribution over $\mathcal{G}_D$) $b_t : \mathcal{G}_D \rightarrow [0, 1]$, with $\sum_g b_t(g) = 1$.

3.2 Agent D's Objective: Path vs. Deception

Agent D solves a **weighted multi-objective** problem:

$$\xi_D^* = \arg \min_{\xi_D \in \Xi_D} [\alpha J_{\text{path}}(\xi_D) + (1 - \alpha) J_{\text{deception}}(\xi_D)]$$

J_{path} — **path cost**

$$J_{\text{path}}(\xi_D) = \int_0^T \|\dot{\xi}_D(t)\| dt$$

Total length / control effort.

$J_{\text{deception}}$ — **“give-away”**

$$J_{\text{deception}}(\xi_D) = f_{\theta}(\xi_D)(g^*)$$

Probability a **learned classifier** f_{θ} assigns to the **true** goal — we want this **low**.

Trade-off knob $\alpha \in [0, 1]$: 1 $\rightarrow$ optimal path, 0 $\rightarrow$ max deception.

Solver. Adversarial RRT* (uses the combined cost above as its rewire metric).

Inference (offline $\rightarrow$ online)

Offline — Inverse RL. Given expert demonstrations of D , recover a reward $R_D(s, a)$ that explains them. Plain English: “given good behavior, infer the cost that produced it.”

Online — Particle Filter. Maintain N weighted goal-hypotheses; re-weight each by Bayes’ rule using R_D as the likelihood:

$$b_{t+1}(g) \propto P(\xi_D[t, t+1] \mid g) b_t(g)$$

MAP estimate: $\hat{g}(t) = \arg \max_g b_t(g)$.

Planning — Game-Theoretic MPC

At every timestep, solve a **finite-horizon** optimization:

$$\min_{\xi_I[t, t+H]} J_I(\xi_I, \hat{\xi}_D)$$

$\hat{\xi}_D$ = belief-weighted predicted trajectory of D .

$$J_I = \|\xi_I(T_{\text{int}}) - \hat{g}\|^2 + \int_t^{t+H} \|\dot{\xi}_I\|^2 d\tau$$

Execute one step, **replan**, repeat.

A **two-player dynamic game with asymmetric information**:

- Agent D knows g^* ; Agent I does **not**.
- **Stackelberg-style**. D leads, I responds to its observations.

Termination conditions:

- **Agent D wins**: $\|x_D(t) - g^*\| < 0.5$ (reaches its true goal).
- **Agent I wins**: $\|x_I(t) - x_D(t)\| < 0.5$ (catches D).
- **Timeout** at 60 s.

Evaluated empirically via simulation (no closed-form equilibrium).

4. Implementation

4.1 System Architecture

Stack. Python + JAX + Equinox — automatic differentiation, GPU acceleration, pytree-based trajectory representation.

Agent D pipeline.

- GRU recurrent network $\rightarrow$ surrogate observer
- Deception cost $f_{\theta}(\xi)(g^*)$
- Adversarial RRT* (rewires by combined cost)

Agent I pipeline.

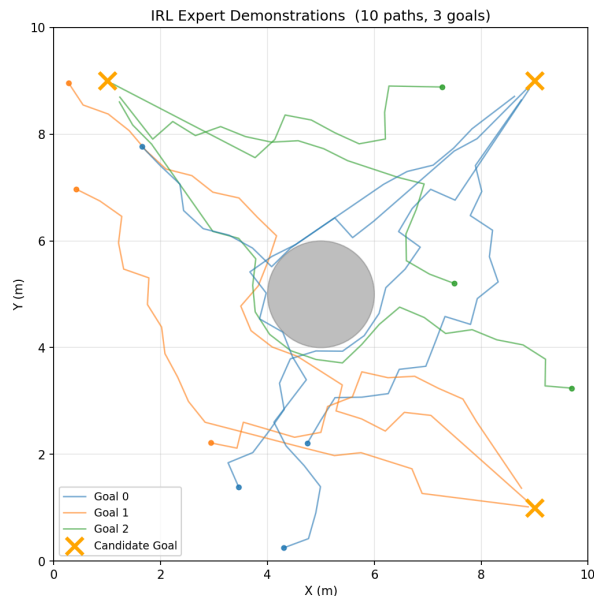
- 3-layer MLP $\rightarrow$ reward R_D (contrastive IRL)
- Particle filter (200 particles) $\rightarrow$ belief tracker
- Adam-based gradient-descent MPC

Two-stage pipeline.

- **Offline:** train both networks on synthetic demonstrations, serialize weights to disk.
- **Online:** load checkpoints, run adversarial sim, log trajectories + metrics + figures.

Performance details.

- Observer input truncated to last 20 positions to avoid JIT recompiles on growing arrays.
- RRT* rewiring uses path-cost only (skip the per-neighbor observer call).



Training input for Agent I's reward network.

- 200 **expert paths** generated by a **goal-directed RRT** — no deception, just direct routing toward one of the three candidate goals.
- Color-coded by target goal (10 paths/goal shown).
- Demonstrations start from random positions and respect the central-obstacle collision constraint.
- **Contrastive IRL** uses these: for each (s, a) in a demo, push the learned reward R_D above 8 random alternative actions.

Setup. 10×10 workspace, single circular obstacle (radius 1) at (5, 5). Goals at (9, 9), (9, 1), (1, 9). D starts (1, 1), I starts (1, 9); true goal $g^* = (9, 9)$.

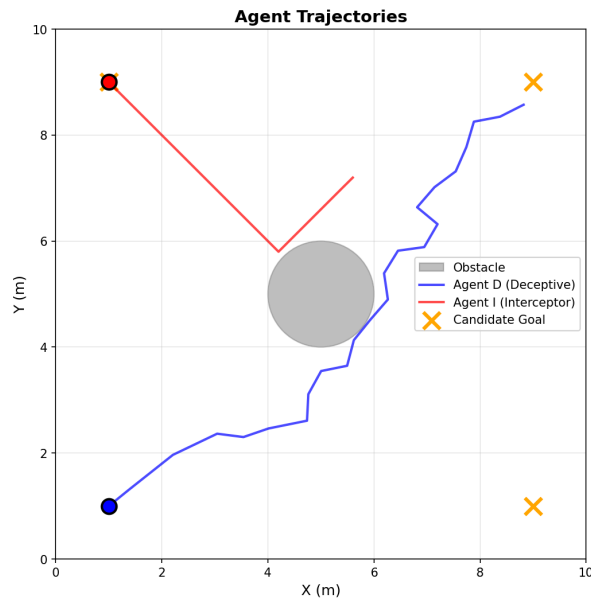
Primary trial.

- $\alpha = 0.5$
- 2000 Adv. RRT* iters, step 0.5
- Particle filter: 200 particles
- MPC: horizon 15, 50 grad steps
- $\Delta t = 0.1$ s

Ablation. (planning-only)

- $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$
- 300 RRT* iters per α
- Same observer reused
- Measures: obs. accuracy, path ratio

5. Results

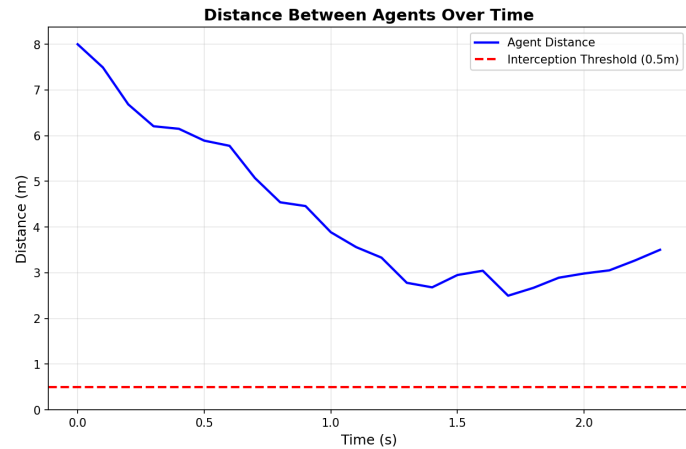


Outcome. Agent D wins at $t = 2.30$ s.

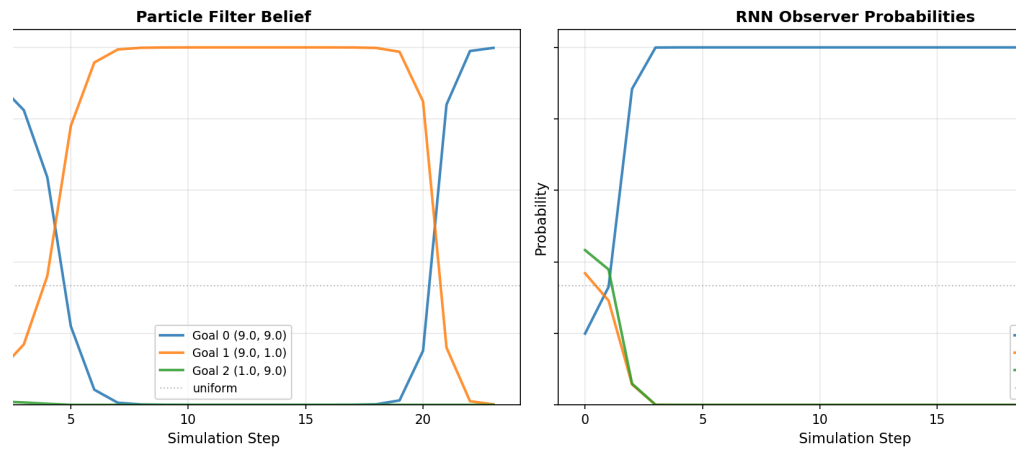
Path length ratio. $J_{\text{path}}(\xi_D^*)/J_{\text{path}}(\xi_D^{\text{opt}}) = 1.124$ — about 12% above optimal (a modest detour).

Interception. Agent I never closes within the 0.5-unit threshold.

Note: The deceptive trajectory swings northwest before committing to (9, 9).

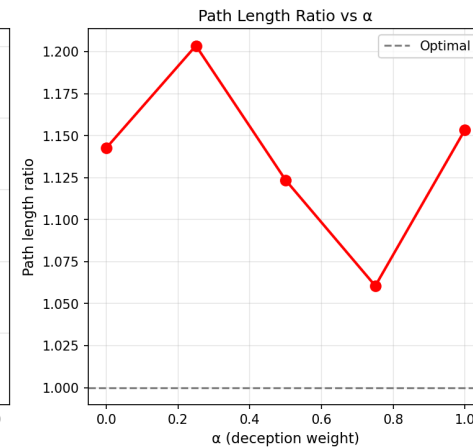
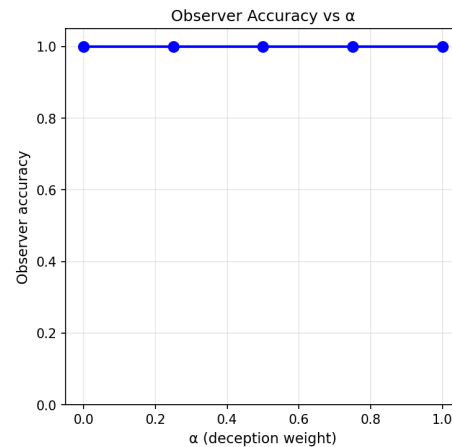
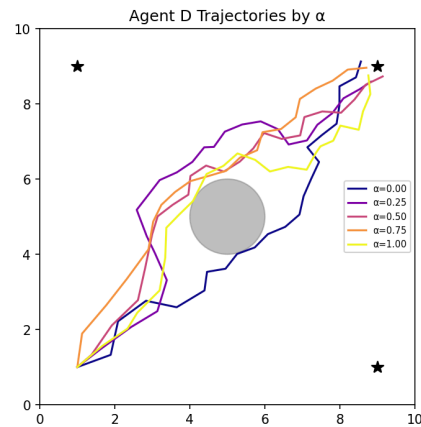


Distance. Min approach 2.50 units — **5x** the 0.5-unit interception threshold.



Belief.

- **Particle filter:** converges to true goal in ≈ 1.0 s.
- **RNN observer:** assigns probability **1.000** from frame 1.



Observer accuracy: 1.000 for every $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$.

Path length ratio: 1.06 – 1.20, no monotone trend with α .

Implication. Tuning α buys you **zero** additional deception in this workspace — every dollar spent on detours is **wasted**. The deception layer is effectively **inert**.

6. Discussion

Goal-set geometry dominated outcomes, not the deception parameterization.

Goal layout matters:

- Three **corner** goals → every trajectory commits to a quadrant within a few steps.
- The RNN observer saturates at perfect classification **regardless of α** .
- Contrast: Nichols et al. arrange goals on a **common arc** — long shared prefix, deception works ($\approx 10\%$ observer accuracy).

Interceptor failure mode:

- Belief converges **correctly** in 1 s, but...
- MPC horizon (15 steps) + short episode (2.3 s) → I can't physically catch up in time.
- IRL training cut loss 56%, but the per-step likelihoods weren't sharp enough to drive aggressive interception.

7. Conclusion

7.1 Summary and Takeaways

Built (JAX/Equinox).

- Agent D: GRU observer + Adversarial RRT*.
- Agent I: contrastive IRL + particle filter + game-theoretic MPC.

Found. Agent D **won** (2.30 s, $\approx 12\%$ over optimal); but the deception layer was **inert** — observer accuracy 1.000 for **every** α tested.

Takeaway. Goal-set geometry is a first-class **design variable** in adversarial motion-planning evaluation.

Future work. Ambiguous goal layouts, 3D, randomized trials, max-entropy IRL, longer MPC horizon, closed-loop D replanning.

References